

Sound Synthesis Models

A Complete Visual Guide for Beginners

This guide walks you through all 6 major sound synthesis methods — how each one works, why it was invented, and what makes it different. No music theory required.

#	Model	Core idea	Best for
1	Additive Synthesis	Building sound by layering simple waves	Organs, bells, choirs, glassy pads
2	Subtractive Synthesis	Carving sound from a rich, buzzy wave	Bass lines, leads, pads, classic synth sounds
3	FM Synthesis	One wave bending another to create complexity	Electric pianos (DX7), bells, glassy metallics, xylophone, classic 80s sounds
4	Wavetable Synthesis	Morphing between stored wave snapshots	Modern EDM, evolving pads, leads, any sound that morphs
5	Granular Synthesis	Scattering tiny sound grains into new textures	Cinematic atmospheres, time-stretching, voice morphing, glitchy textures, drone pads
6	Physical Modelling	Simulating real physics to create natural sound	Acoustic piano (Pianoteq), guitar, flute, bowed strings, vocals — expressive realism

Model 1 of 6

Additive Synthesis

Building sound by layering simple waves

The simple idea

All sounds in nature are made of many simple waves mixed together. Additive synthesis does exactly that — it stacks sine waves (the simplest wave shape) on top of each other to build any sound you want.

Why it was invented

Scientists discovered (Fourier, 1800s) that every sound — a violin, a voice, a bell — can be broken into simple sine waves. So if you reverse that, you can build any sound by combining them. It is the most mathematically honest way to make sound.

How it works — step by step

1 Start with Wave 1

A pure sine wave — the smoothest, simplest sound (like a flute note).

2 Add Wave 2 (double frequency)

This second wave is twice as fast. Together they start sounding richer.

3 Add Wave 3, 4, 5...

Each extra wave (called a harmonic) adds more texture and character.

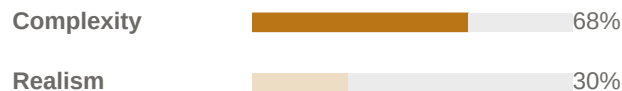
4 Adjust volumes

By making each harmonic louder or quieter, you shape the final sound.

Used for

Organs, bells, choirs, glassy pads. Any sound with clear harmonic structure.

Complexity vs realism



Model 2 of 6

Subtractive Synthesis

Carving sound from a rich, buzzy wave

The simple idea

Instead of building up from nothing, start with a very rich, buzzy sound that contains all frequencies — then use a filter to carve away what you do not want. Think of it like a sculptor chiselling stone.

Why it was invented

It is far more intuitive and practical than additive. Instead of adding 50+ waves precisely, you start rich and just cut. This is why it became the dominant method in classic analogue synthesizers (Moog, Prophet-5) from the 1960s to 1980s.

How it works — step by step

- 1

Oscillator makes a raw wave
A sawtooth or square wave — harsh, buzzy, full of energy across all frequencies.
- 2

Filter cuts frequencies
A low-pass filter removes high frequencies above your cutoff point. Sounds softer.
- 3

Envelope shapes the volume
An ADSR envelope controls how the volume rises and falls over time.
- 4

LFO adds movement
A slow wobble (LFO) makes the filter or pitch move, giving life to the sound.

Used for

Bass lines, leads, pads, classic synth sounds. The backbone of most electronic music.

Complexity vs realism



FM Synthesis

One wave bending another to create complexity

The simple idea

FM stands for Frequency Modulation. One wave (the modulator) wobbles the frequency of another wave (the carrier) very rapidly. This creates complex, metallic, digital-sounding tones from very little.

Why it was invented

John Chowning invented it at Stanford in 1967. It creates sounds rich with harmonics using just two simple waves — far fewer resources than additive. Yamaha licensed it for the DX7 (1983), one of the best-selling synthesizers ever.

How it works — step by step

- 1

Carrier wave plays the note
A simple sine wave at the pitch you want to hear.
- 2

Modulator controls frequency
A second sine wave rapidly speeds up and slows down the carrier.
- 3

Modulation index controls depth
Low = gentle shimmer. High = harsh metallic clang.
- 4

Multiple FM pairs (operators)
Stack more carrier-modulator pairs for even richer results.

Used for

Electric pianos (DX7), bells, glassy metallics, xylophone, classic 80s sounds.

Complexity vs realism



Model 4 of 6

Wavetable Synthesis

Morphing between stored wave snapshots

The simple idea

Record one single cycle of any waveform as a snapshot — then play that snapshot on a loop at any speed (pitch). Store many snapshots in a table and smoothly morph between them for evolving, shifting sounds.

Why it was invented

Wolfgang Palm invented it in 1978 (PPG Wave). The breakthrough: capture the character of a real instrument's wave shape and reproduce it at any pitch — without a full sample recording. Modern synths like Serum, Vital, and Massive are all wavetable.

How it works — step by step

- 1

Capture a single-cycle waveform
Record or design one cycle — sine, saw, complex, or real instrument.
- 2

Store many cycles in a table
The table holds dozens to hundreds of different wave shapes in sequence.
- 3

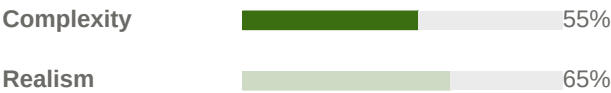
Loop the cycle at the right speed
Play it at exactly the needed pitch by looping faster or slower.
- 4

Scan / morph through the table
Move through the table over time for evolving, morphing textures.

Used for

Modern EDM, evolving pads, leads, any sound that morphs. Industry standard today.

Complexity vs realism



Model 5 of 6

Granular Synthesis

Scattering tiny sound grains into new textures

The simple idea

Take any sound. Chop it into tiny pieces called grains — each only 1 to 100 milliseconds long. Then scatter, overlap, stretch, and rearrange those grains to create entirely new textures. Like shredding a photo and rearranging the scraps.

Why it was invented

Iannis Xenakis and Dennis Gabor proposed the theory in the 1940s–50s, based on quantum physics. The key insight: sound is a series of tiny quanta. Controlling each grain independently lets you stretch time without changing pitch — impossible with basic playback.

How it works — step by step

- 1

Load any source sound
A voice, a string, a field recording — anything works as a source.
- 2

Slice into grains
The engine cuts the sound into tiny 1–100ms fragments.
- 3

Scatter and overlap grains
Grains are triggered rapidly, often overlapping, with random variation.
- 4

Control parameters
Grain size, density, pitch, position, randomness — each changes the texture.

Used for

Cinematic atmospheres, time-stretching, voice morphing, glitchy textures, drone pads.

Complexity vs realism



Physical Modelling

Simulating real physics to create natural sound

The simple idea

Instead of recording or approximating sounds, use physics equations to simulate exactly how a real object vibrates. A guitar string, a flute tube, a drumhead — all modelled with mathematics in real time.

Why it was invented

Julius O. Smith developed Karplus-Strong (1983) and waveguide synthesis in the 1980s–90s. Samples only capture one specific hit. Physical models react dynamically — you can change bow pressure or breath while playing, just like a real instrument.

How it works — step by step

- 1

Define the physical object
Set material, size, shape, tension — a string, tube, or membrane.
- 2

Apply excitation
A pluck, bow, breath, or strike — the energy input that starts vibration.
- 3

Simulate wave propagation
Physics equations calculate how vibrations travel and reflect.
- 4

Model resonance and decay
Energy loss, damping, and resonance are all calculated naturally.

Used for

Acoustic piano (Pianoteq), guitar, flute, bowed strings, vocals — expressive realism.

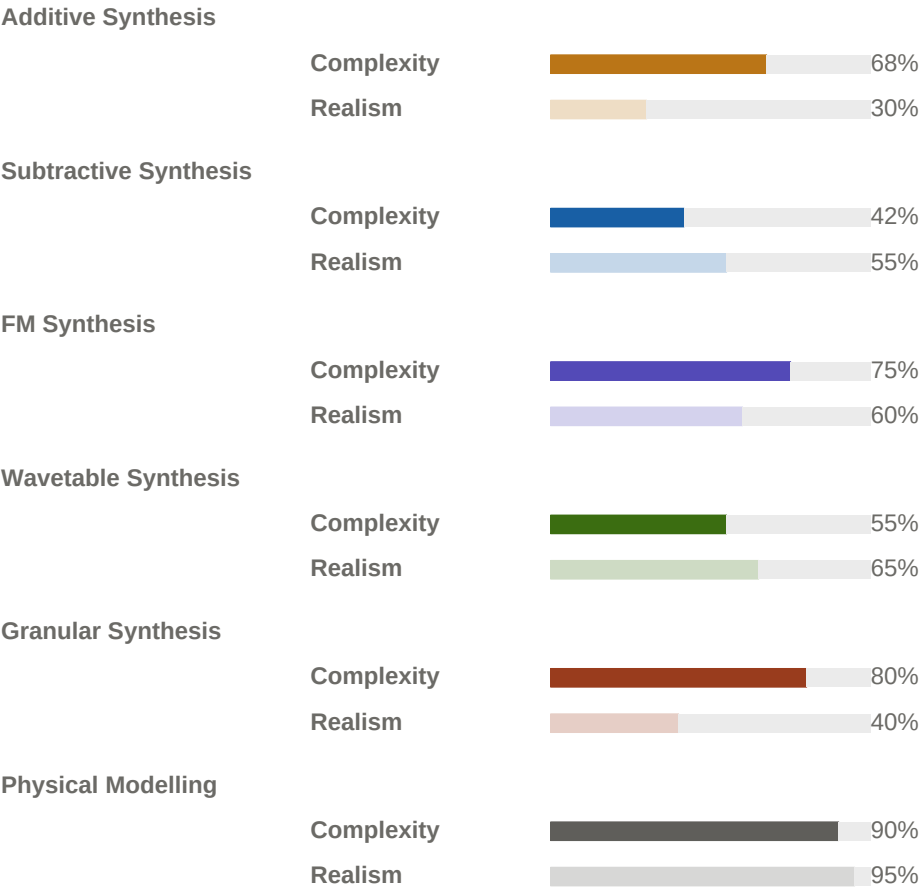
Complexity vs realism



Compare All Models

See how every model stacks up at a glance.

Complexity vs Realism



Full comparison table

Model	Core idea	Best for	CPU cost
Additive	Layer sine waves	Organs, bells	Medium
Subtractive	Filter a rich wave	Bass, leads, pads	Low
FM	Wave modulates wave	80s digital, e-piano	Low
Wavetable	Morph wave snapshots	Modern EDM, leads	Low–Med
Granular	Scatter tiny grains	Textures, time-stretch	High
Physical Modelling	Simulate physics	Acoustic instruments	Very High

Key differences in one sentence each

Additive vs Subtractive

Additive builds up from silence. Subtractive starts rich and carves away.

Subtractive vs Wavetable

Subtractive uses maths-generated waves. Wavetable uses captured real wave shapes.

FM vs Additive

Both create harmonics, but FM does it with just 2 waves. Additive needs dozens.

Granular vs Wavetable

Wavetable loops one cycle perfectly. Granular chops any real sound chaotically.

Physical vs all others

All others approximate. Physical models calculate sound from actual physics.